ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 0.68 m and 3 m	lambda = v/f.
3	**(c)** 24 m/s	0.6 x 40.
4	**(d)** 150 m	1500 x 0.2/2.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 2 s	680/340.
7	**(a)** 86 m	vt.
8	**(a)** 340 m/s	34/0.1 = 340 - note: for illustration.
9	**(d)** A is false but R is true.	A is false (loudness does not depend on frequency); R is true.
10	**(c)** 1 : 3	Distance proportional to time.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance = 170 m (1); v = 170/0.5 = 340 m/s (1).	1+1
12	Different harmonic content (1); Waveform shapes differ (1).	1+1
13	Sound travels faster in steel (1); It reaches the ear earlier (1).	1+1
14	t steel = 0.2 s (1); t air = 2.94 s; gap 2.74 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wavelengths 4 : 1 (1); Speeds equal (1); Loudness equal; pitch higher for 400 Hz (1).	1+1+1
16	T = 0.00143 s (1); lambda = 0.486 m (1); Waves in 5 s = 3500 (1).	1+1+1
17	Reflections from the walls prolong it (1); Reverberation (1); No reflections in the open (1).	1+1+1
	OR: d1 = 340 m (1); d2 = 510 m (1); Distance between cliffs = 850 m (1).	1+1+1
18	Initial distance 510 m (1); New distance 425 m (1); New echo time 2.5 s (1).	1+1+1
	OR: Bell rings but no sound in vacuum (1); Microphone converts sound to electrical signals (1); Speaker converts back (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	lambda = 0.782 m (1); t = 344/344 = 1 s (1); Waves between source and wall = 172/0.782 = 220 (1); Echo path 344 m: 440 waves (1); Frequency equals the waves per second (1).	1+1+1+1+1
	OR: Time gap less than 0.1 s in a small room (1); A big hall delays reflections (1); Use of absorbers (1); Curved ceilings and sound boards (1); Ceiling height above 17 m avoided or treated (1).	1+1+1+1+1
20	Observer sees flash/clap and starts stopwatch (1); Stops when the sound is heard (1); v = 1020/3 = 340 m/s (1); Repeat and take the mean (1); Precautions: large distance, wind, reaction time (1).	1+1+1+1+1
	OR: SONAR: depth and fish (1); Sonography: foetus, liver (1); Testing: cracks in metals (1); Limitations: absorption and resolution (1); Safer than X-rays (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 20 m	t1 = sqrt(2h/g), t2 = h/v: h = 20 gives 2 + 0.06.
2	**(c)** the speed of sound in steel is greater	Faster in solids.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** 3.13 s	3 s + 0.13 s.
5	**(d)** 2 : 1	f = v/lambda.
6	**(c)** 1478 m/s	340/0.23.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(d)** 3 m	lambda = v/f.
9	**(d)** not hear it as it is infrasonic	Below 20 Hz.
10	**(d)** Speed is higher at 34 degrees Celsius	Same distance, less time.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	d = vt/2 (1); = 300 m (1).	1+1
12	d = 340 x 8 = 2720 m (1); Light takes about 10 microseconds (1).	1+1
13	Speed is higher in helium (1); Same cavity size, so higher frequency (1).	1+1
14	Distance is small (1); Echo returns within 0.1 s and merges with the original (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Depth = 2772 m (1); Shoal depth = 693 m (1); Height above the bed = 2079 m (1).	1+1+1
16	20 Hz: 1200 per minute; 20 kHz: 1200000 (1); lambda 17 m and 1.7 cm (1); Range of hearing (1).	1+1+1
17	d1 = 1540 x 10 x 10^-6/2 = 7.7 mm (1); d2 = 9.24 mm (1); Thickness = 1.54 mm (1).	1+1+1
	OR: lambda1 = 1.344 m (1); lambda2 = 1.367 m (1); Frequency unchanged (1).	1+1+1
18	Cooler air near the ground (1); Sound refracts down (1); Less noise in the background (1).	1+1+1
	OR: In speaking, vibrations reach the ear through the bones of the skull as well (1); Recording carries sound only through air (1); Different transmission changes the quality (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Iceberg distance = 4350 m (1); Sea bed depth = 5800 m (1); Difference = 1450 m (1); d = vt/2 for each (1); Echo from nearer surface returns first (1).	1+1+1+1+1
	OR: T = 0.00167 s (1); Air lambda = 0.567 m (1); Water lambda = 2.5 m (1); 600 waves per second in both (1); Frequency is unchanged (1).	1+1+1+1+1
20	Flute: high pitch, steady (1); Drum: low pitch, decays (1); Harder strike increases amplitude (1); Loudness increases, pitch unchanged (1); Quality differs due to overtones (1).	1+1+1+1+1
	OR: d1 = 340 x 0.6/2 = 102 m (1); d2 = 170 m (1); Distance between the walls = 272 m (1); At the middle each wall is 136 m away (1); Echo time = 272/340 = 0.8 s (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 345.6 m/s	f lambda.
3	**(d)** 1 m	v/f.
4	**(d)** the medium is removed	Sound needs a medium.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R describes sound but does not explain the light's nature.
6	**(d)** 2 : 1	lambda proportional to 1/f.
7	**(d)** at the upper limit of human hearing	Limit.
8	**(a)** 1000 Hz	v/lambda.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** 2 m	lambda/2 = 4/2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	T = 0.00227 s (1); lambda = 0.77 m; 880 waves (1).	1+1
12	In water 1.875 m (1); In air 0.425 m; ratio 4.4 (1).	1+1
13	lambda = 5.7 mm (1); Short wavelength detects small insects and gives directional waves (1).	1+1
14	Initial distance 34 m (1); New 17 m; time 0.1 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No medium (1); Light and radio waves carry information (1); Sound needs particles (1).	1+1+1
16	Pulse reflects from the back (1); t = 2d/v (1); d = 5000 x 4 x 10^-6/2 = 0.01 m (1).	1+1+1
17	Frequency ranges (1); Sources: human voice, earthquakes, bats (1); Uses: communication, prediction, imaging (1).	1+1+1
	OR: Table vibrates with the fork (1); A larger surface sets more air in vibration (1); Louder sound for a shorter time (1).	1+1+1
18	Air: 3 s (1); Steel: 0.204 s (1); Gap 2.8 s (1).	1+1+1
	OR: Intensity falls as 1/d squared (1); Doubling distance gives one-fourth intensity (1); Amplitude decreases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Shorter string: higher frequency (1); More tension: higher frequency (1); Thinner string: higher frequency (1); Thick strings give lower notes (1); Tuning pegs adjust tension (1).	1+1+1+1+1
	OR: Microphone: sound to electrical signal (1); Amplifier increases the signal (1); Loudspeaker: electrical to sound by a vibrating cone (1); Sound spreads and is heard (1); Energy conversions and losses (1).	1+1+1+1+1
20	Time = 34/340 = 0.1 s (1); lambda = 5.7 mm (1); Distance from time delay (1); Size from the strength of echo (1); Direction from the two ears (1).	1+1+1+1+1
	OR: Values 331, 337, 343, 349, 355 (1); Straight line graph (1); Slope 0.6 m/s per degree Celsius (1); Intercept 331 m/s (1); Used in echo calculations (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.